

CompTIA A+ Acronyms (Core 1 and 2) 1201/1202

#	Acronym	Expansion	Definition
1	AAA	Authentication, Authorization, and Accounting	Security framework that controls access by verifying identity (authentication), deciding permissions (authorization), and tracking activity (accounting).
2	ACL	Access Control List	A rule list on devices or file systems that allows or denies specific traffic or user actions.
3	ADF	Automatic Document Feeder	Printer/scanner feeder that automatically pulls multiple pages for scanning or copying.
4	AES	Advanced Encryption Standard	Widely used symmetric-key encryption standard that protects data at rest and in transit (e.g., WPA2/WPA3).
5	AMD	Advanced Micro Devices, Inc.	Hardware maker of PC processors and graphics chips used in desktops and laptops.
6	AP	Access Point	Wireless access point; bridges Wi-Fi clients to the wired LAN.
7	APFS	Apple File System	Apple's modern file system with features like snapshots, cloning, and strong encryption.
8	APIPA	Automatic Private Internet Protocol Addressing	Windows feature that self-assigns a 169.254.x.x address when DHCP is unavailable.
9	ARM	Advanced RISC [Reduced Instruction Set Computer] Machine	CPU architecture optimized for low power; common in phones, tablets, and some laptops.
10	ATX	Advanced Technology Extended	Motherboard and power-supply form factor standard that defines size, mounting, and connectors.
11	AUP	Acceptable Use Policy	Policy that sets rules for acceptable use of an organization's computers, network, and Internet access.
12	BEC	Business Email Compromise	Social-engineering scam where attackers impersonate executives to trick staff into sending money or data.
13	BIOS	Basic Input/Output System	Firmware that initializes hardware and starts the boot process on older PCs (superseded by UEFI on newer systems).
14	BNC	Bayonet Neill-Concelman	Bayonet-style RF coax connector used on older Ethernet and video equipment.
15	BSOD	Blue Screen of Death	Windows stop error screen indicating a critical system failure requiring reboot and troubleshooting.
16	BYOD	Bring Your Own Device	Practice of employees using personal devices for work; requires strong security controls and policy.
17	CAC	Calling-card Authorization Computer	Common Access Card: a smart ID card providing strong authentication, commonly used in government/DoD environments.
18	CAS	Column Address Strobe	Signal used by RAM to locate and access a specific column of data. It works with RAS (Row Address Strobe) to identify exact memory locations. CAS latency (CL) is a key performance metric lower is faster.
19	CIFS	Common Internet File System	File-sharing protocol (early form of SMB) used to access files and printers over a network.

20	CMDB	Configuration Management Database	Repository that tracks IT assets, configurations, and their relationships to support change/incident management.
21	CMOS	Complementary Metal-Oxide Semiconductor	Battery-backed chip that stores BIOS/UEFI settings and maintains the real-time clock.
22	CNAME	Canonical Name	DNS record that creates an alias from one hostname to another (canonical) name.
23	CPU	Central Processing Unit	The central processor that performs calculations and runs instructions for the system.
24	DB-9	Serial Communications D-Shell Connector, 9 pins	9-pin D-sub serial connector often used for RS-232 on legacy PCs and network gear.
25	DDOS	Distributed Denial of Service	Distributed Denial of Service: a technology or concept used in IT; know what it is and when to use it for the A+ exam.
26	DDR	Double Data Rate	Type of synchronous DRAM that transfers data on both clock edges; used as main system memory.
27	DHCP	Dynamic Host Configuration Protocol	Service that automatically assigns IP addresses and network settings to clients.
28	DIMM	Dual In-line Memory Module	Removable memory module that plugs into motherboard slots to add RAM.
29	DKIM	DomainKeys Identified Mail	Email authentication using cryptographic signatures to verify messages weren't altered and are authorized by the sending domain.
30	DLP	Data Loss Prevention	Tools and policies that prevent sensitive data from leaving the organization (email, web, storage).
31	DMARC	Domain-based Message Authentication, Reporting, and Conformance	Email policy telling receivers how to handle messages failing SPF/DKIM and providing domain-owner reporting.
32	DNS	Domain Name System	System that translates human-readable domain names into IP addresses.
33	DOS	Denial of Service	Denial of Service: a technology or concept used in IT; an attack that overwhelms a server, service, or network with a flood of internet traffic from multiple sources.
34	DRM	Digital Rights Management	Technologies that control how digital content can be accessed, copied, or shared.
35	DSL	Digital Subscriber Line	Internet service that uses telephone lines to provide broadband connectivity.
36	DVI	Digital Visual Interface	Digital video interface standard for connecting computers to monitors.
37	ECC	Error-correcting Code	Error-correcting Code memory that adds extra bits so the system can detect and correct single-bit errors, used for higher reliability (often in servers).
38	EDR	Endpoint Detection and Response	Endpoint Detection and Response security that continuously monitors endpoint activity, detects suspicious behavior, and helps contain or remediate threats.
39	EFS	Encrypting File System	Encrypting File System, a Windows feature that encrypts files/folders on NTFS volumes so only authorized users can read them.
40	EOL	End-of-life	End-of-Life, when a vendor stops updates and support for a product; using EOL gear increases security and reliability risk.
41	eSATA	External Serial Advanced Technology Attachment	External Serial ATA, an external connector for SATA storage that provides fast, direct-attach disk performance (largely replaced by USB 3.x/USB-C).
42	ESD	Electrostatic Discharge	Electrostatic Discharge, static electricity that can damage components; controlled with wrist straps, mats, and proper handling.

43	eSIM	Embedded Subscriber Identity Module	Built-in digital SIM inside a mobile device that allows you to activate cellular service without inserting a physical SIM card. Common in newer smartphones and IoT devices
44	EULA	End-user License Agreement	End-User License Agreement, the legal terms that define how you may install and use software.
45	exFAT	Extended File Allocation Table	Extended File Allocation Table, a Microsoft file system optimized for flash drives; supports very large files and is widely compatible across OSes.
46	FaaS	Function as a Service	Cloud computing model where developers run code in response to events without managing servers. Examples include serverless platforms like AWS Lambda, Azure Functions. You only pay when the function runs.
47	FAT32	32-bit File Allocation Table	32-bit File Allocation Table, an older, very compatible file system for removable media; limited to 4 GB per file and 2 TB volumes.
48	FRT	Facial Recognition Technology	Facial Recognition Technology, biometric tech that identifies or verifies a person using their face (used for device unlock and access control).
49	FTP	File Transfer Protocol	File Transfer Protocol, legacy file transfer protocol (TCP/21) that sends data in clear text; prefer SFTP/FTPS for security.
50	GFS	Grandfather-Father-Son	Grandfather-Father-Son, a backup rotation scheme using daily (son), weekly (father), and monthly (grandfather) backup sets.
51	GPS	Global Positioning System	Global Positioning System, satellite-based navigation and time source used for location services and accurate timekeeping.
52	GPT	GUID [Globally Unique Identifier] Partition Table	GUID Partition Table, modern disk partitioning standard that supports >2 TB drives and many partitions; required by UEFI systems.
53	GPU	Graphics Processing Unit	Graphics Processing Unit, processor specialized for rendering graphics and accelerating parallel workloads (e.g., video, AI).
54	GUI	Graphical User Interface	User interface that allows interaction with a device using visual elements like windows, icons, buttons, and menus instead of text commands. Examples include Windows and macOS desktops.
55	GUID	Globally Unique Identifier	Globally Unique Identifier, a 128-bit value used to uniquely identify objects (files, partitions, software components) across systems.
56	HD	High Definition	High Definition, video/display resolution such as 720p/1080p that provides sharper images than standard definition.
57	HDD	Hard Disk Drive	Hard Disk Drive, magnetic spinning-platter storage offering large capacity at low cost but slower than SSDs and more shock-sensitive.
58	HDMI	High-definition Media Interface	High-Definition Multimedia Interface, digital audio/video interface for displays and TVs; carries video and multi-channel audio over one cable.
59	HSM	Hardware Security Module	Hardware Security Module, tamper-resistant device that generates, stores, and uses cryptographic keys securely.
60	HTTP	Hypertext Transfer Protocol	Hypertext Transfer Protocol, the basic web protocol (TCP/80) used for transferring web pages and APIs without encryption.
61	HTTPS	Hypertext Transfer Protocol Secure	HTTP Secure, HTTP over TLS/SSL (TCP/443) that encrypts and authenticates web traffic to protect data in transit.

62	IaaS	Infrastructure as a Service	Infrastructure as a Service, cloud model providing virtual machines, storage, and networking so you don't manage physical hardware.
63	IAM	Identity and Access Management	Identity and Access Management, policies and tools to authenticate users and control what resources they can access.
64	IMAP	Internet Mail Access Protocol	Internet Message Access Protocol, email protocol (TCP/143; SSL/TLS 993) that keeps mail on the server and syncs across devices.
65	IOPS	Input/Output Operations Per Second	Input/Output Operations Per Second, storage performance metric indicating how many reads/writes a device can handle each second.
66	IoT	Internet of Things	Internet of Things, networked devices (sensors, appliances, cameras) that collect and exchange data; often require strong security.
67	IP	Internet Protocol	Internet Protocol, the addressing and routing system (IPv4/IPv6) that delivers packets across networks.
68	IPS	In-plane Switching	In-Plane Switching, LCD display panel technology known for better color accuracy and wide viewing angles compared to TN/VA.
69	IR	Infrared	Infrared, short-range, light-based wireless communication used in remote controls and some proximity sensors.
70	ISO	International Organization for Standardization	International Organization for Standardization, develops global standards; in IT you'll also see "ISO image," a single-file copy of an optical disc.
71	ISP	Internet Service Provider	Company that provides internet access to users. Examples include Comcast, AT&T, and Spectrum. They connect your home or business network to the internet.
72	ITX	Information Technology eXtended	Information Technology eXtended, small motherboard form-factor family (e.g., Mini-ITX) used in compact PC builds.
73	KVM	Keyboard-Video-Mouse	A hardware switch or device that lets you control multiple computers using a single keyboard, video display, and mouse.
74	LAN	Local Area Network	A local network that connects computers and devices within a small area (home, office, school) for sharing data and resources.
75	LC	Lucent Connector	A small form-factor fiber-optic connector that uses a latch; common for high-density network patch panels.
76	LCD	Liquid Crystal Display	Flat-panel display technology that uses liquid crystals and a backlight; common in monitors and laptops.
77	LDAP	Lightweight Directory Access Protocol	A network protocol for querying and managing directory services (users, groups, permissions) such as Active Directory.
78	LED	Light-emitting Diode	Light-emitting diode technology used for indicator lights and as a backlight or pixel source in modern displays.
79	LTE	Long-Term Evolution	A 4G cellular data standard that provides high-speed wireless Internet over mobile networks.
80	MAC	Media Access Control	A unique hardware address assigned to a network interface used for Layer 2 communication on a LAN.
81	MAN	Metropolitan Area Network	A network that spans a city or campus area, larger than a LAN but smaller than a WAN.

82	MBR	Master Boot Record	Legacy disk partitioning and boot sector format used by BIOS systems; stores the bootloader and partition table.
83	MDM	Mobile Device Management	Software suite that centrally manages and secures mobile devices (enforce passcodes, push configs, remote wipe).
84	MDR	Managed Detection and Response	A managed security service that provides continuous threat monitoring, detection, and response by a provider.
85	MFA	Multifactor Authentication	An authentication method requiring two or more verification factors (e.g., password + app code or token).
86	MFP	Multifunction Printer	A multi-function printer that combines printing, scanning, copying, and often faxing in one device.
87	MMC	Microsoft Management Console	Microsoft Management Console, Windows framework that hosts administrative snap-ins (e.g., Device Manager).
88	MNDA	Mutual Non-disclosure Agreement	Mutual Non-Disclosure Agreement, both parties agree to keep shared confidential information private.
89	mSATA	Mini-serial Advanced Technology Attachment	A compact SATA-based SSD form factor used in small devices and older laptops; connects via SATA interface.
90	MSDS	Material Safety Data Sheet	Material Safety Data Sheet (now SDS), explains hazards and safe handling of chemicals like toner or cleaners.
91	MX	Mail Exchange	A DNS record type that specifies which mail servers accept email for a domain.
92	NAC	Network Access Control	Network Access Control, checks a device's security posture and controls whether it can join the network.
93	NAS	Network Attached Storage	Dedicated storage device connected to a network that allows multiple users/devices to store and access files centrally. Common in homes and businesses for file sharing and backups
94	NAT	Network Address Translation	Method used by routers to translate private IP addresses into a public IP address so devices on a local network can access the internet. Helps conserve IP addresses and adds a layer of security
95	NDA	Non-disclosure Agreement	Non-Disclosure Agreement, a legal contract to keep specified information confidential.
96	NetBIOS	Network Basic Input/Output System	Legacy Windows networking API/protocol suite for name services and sessions within a LAN.
97	NFC	Near-field Communication	Near Field Communication, very short-range wireless used for tap-to-pay, pairing, and badge access.
98	NIC	Network Interface Card	Network Interface Card/Controller, the hardware that connects a computer to a network (wired or wireless).
99	NTFS	New Technology File System	Windows file system that supports permissions, encryption, compression, quotas, and large volumes.
100	NTP	Network Time Protocol	Network Time Protocol, keeps computer clocks synchronized over a network.
101	NVMe	Non-volatile Memory Express	High-performance storage protocol for SSDs that uses PCIe lanes for very fast reads/writes.
102	OEM	Original Equipment Manufacturer	Company that produces hardware or software that is sold under another company's brand. In IT, OEM licenses are tied to specific hardware (like Windows preinstalled on a PC).
103	OLED	Organic Light-emitting Diode	Display tech where each pixel emits its own light; offers deep blacks and high contrast.

104	ONT	Optical Network Termina	Optical Network Terminal, carrier device that converts fiber signals to Ethernet at the customer premises.
105	OS	Operating System	Operating System, software that manages hardware and provides services for applications (e.g., Windows, Linux).
106	OTP	One-time Password (or Passcode)	One-Time Password, single-use code (often via app or SMS) used to verify identity during login.
107	PaaS	Platform as a Service	Platform as a Service, cloud service that provides runtime, databases, and tools so you deploy apps without managing servers.
108	PAM	Privileged Access Management	Privileged Access Management, controls, monitors, and secures use of administrative or high-risk accounts.
109	PAN	Personal Area Network	Personal Area Network, short-range network for personal devices like phones, headsets, and wearables.
110	PC	Personal Computer	Personal Computer, a general-purpose computing device for a single user.
111	PCI	Peripheral Component Interconnect	Peripheral Component Interconnect, older expansion bus used for add-in cards inside PCs.
112	PCIe	Peripheral Component Interconnect Express	Peripheral Component Interconnect Express, modern high-speed serial bus used for GPUs, NVMe SSDs, and other cards.
113	PCL	Printer Command Language	Printer Command Language, page description language (from HP) widely supported by printers.
114	PII	Personally Identifiable Information	Personally Identifiable Information, data that can identify a person (e.g., name + SSN, address, DOB).
115	PIN	Personal Identification Number	Personal Identification Number, a short numeric code used to authenticate a user.
116	PIV	Personal Identity Verification	Personal Identity Verification, U.S. federal smart card standard that stores certificates for authentication.
117	PoE	Power over Ethernet	Power over Ethernet, supplies electrical power and data over the same Ethernet cable to devices like APs and cameras.
118	POP	Post Office Protocol	Post Office Protocol (POP3), email retrieval protocol that typically downloads messages to the client.
119	POST	Power-on Self-test	Power-On Self-Test, firmware diagnostics that check hardware components during system startup.
120	PSU	Power Supply Unit	Power Supply Unit, converts AC wall power to regulated DC voltages for PC components.
121	PUP	Potentially Unwanted Program	Potentially Unwanted Program, software that a user didn't intend to install; often adware or bundled tools.
122	PXE	Preboot eXecution Environment	Preboot eXecution Environment, boots a computer from the network to deploy images or run installers.
123	QoS	Quality of Service	Quality of Service, manages and prioritizes network traffic to improve performance for critical applications (e.g., VoIP/video).
124	RADIUS	Remote Authentication Dial-in User Server	Remote Authentication Dial-in User Service, centralized AAA (authentication, authorization, accounting) for network logins and Wi-Fi/VPN access.
125	RAID	Redundant Array of Independent Disks	Redundant Array of Independent Disks, combines multiple drives to improve speed and/or provide fault tolerance depending on RAID level.

126	RAM	Random-access Memory	Random Access Memory, short-term working memory that stores active programs and data for quick CPU access; cleared when power is off.
127	RDP	Remote Desktop Protocol	Remote Desktop Protocol, allows a user to remotely log in to and control a Windows computer over the network.
128	ReFS	Resilient File System	Resilient File System, Microsoft file system with data integrity features, large volume support, and automatic corruption repair.
129	RFID	Radio-frequency Identification	Radio-Frequency Identification, uses radio waves and tags to identify and track items (badges, inventory) without a line of sight.
130	RGB	Red-Green-Blue	Red-Green-Blue, additive color model used by displays and LEDs to create colors by mixing red, green, and blue light.
131	RISC	Reduced Instruction Set Computer	Reduced Instruction Set Computer, CPU architecture using a small set of simple instructions for efficient, high-speed processing used in smartphones, tablets, smartwatches, and embedded systems.
132	RJ11	Registered Jack Function 11	Registered Jack 11, 4P2C/6P2C modular connector commonly used for telephone lines.
133	RJ45	Registered Jack Function 45	Registered Jack 45, 8P8C modular connector typically used for Ethernet network cabling.
134	RMM	Remote Monitoring and Management	Software used by IT administrators or Managed Service Providers (MSPs) to remotely monitor, maintain, and manage clients' computers, servers, and networks.
135	RPM	Revolutions Per Minute	A measurement of rotational speed that indicates how many full turns a mechanical device (like a hard drive spindle or cooling fan) makes each minute.
136	RSR	Rapid Security Response	An urgent security update, typically released outside normal patch cycles, to fix newly discovered vulnerabilities or active threats quickly.
137	SaaS	Software as a Service	A cloud-computing model where applications are hosted online and accessed through a browser rather than installed locally (examples: Microsoft 365, Google Workspace).
138	SAML	Security Assertions Markup Language	An open standard that enables secure exchange of authentication and authorization data between identity providers and service providers, supporting single sign-on (SSO).
139	SAN	Storage Area Network	Storage Area Network, dedicated high-speed network that provides block-level storage to servers.
140	SAS	Serial Attached SCSI [Small Computer System Interface]	Serial Attached SCSI, point-to-point serial protocol for connecting enterprise HDDs/SSDs with high throughput and reliability.
141	SATA	Serial Advanced Technology Attachment	Serial ATA, interface for connecting HDDs/SSDs/ODDs to the motherboard; supports hot-swap and high-speed transfers.
142	SC	Subscriber Connector	Subscriber Connector, push-pull fiber-optic connector with a square body; common in enterprise fiber runs.
143	SCADA	Supervisory Control and Data Acquisition	A system used to monitor and control industrial processes such as utilities, manufacturing, and energy infrastructure.
144	SCSI	Small Computer System Interface	Small Computer System Interface, standard for connecting and transferring data between computers and peripherals (legacy/enterprise).
145	SD	Secure Digital	Secure Digital, family of removable flash memory cards used in cameras, phones, and IoT devices.

146	SDS	Software Defined Storage	Storage architecture where storage resources are managed and controlled by software rather than hardware. It allows flexibility, scalability, and centralized management in cloud and data center environments.
147	SFTP	Secure File Transfer Protocol	SSH File Transfer Protocol, secure file transfer protocol that runs over SSH for encrypted uploads/downloads.
148	SIM	Subscriber Identity Module	Subscriber Identity Module, smart card in mobile devices that stores carrier identity and network credentials.
149	SLA	Service-level Agreement	A formal contract between a service provider and customer that defines performance expectations, uptime guarantees, and response times.
150	S.M.A.R.T	Self-monitoring Analysis and Reporting Technology	A monitoring system built into hard drives and SSDs that detects and reports on drive health to predict potential failures.
151	SMB	Server Message Block	Server Message Block, protocol for Windows file/print sharing and network resource access.
152	SMS	Short Message Service	Standard text-messaging technology used to send and receive short messages on mobile phones.
153	SMTP	Simple Mail Transfer Protocol	Simple Mail Transfer Protocol, protocol used to send outbound email from clients/servers.
154	SNMP	Simple Network Management Protocol	Protocol used to monitor and manage network devices (routers, switches, servers). It collects data like performance metrics, errors, and device status.
155	SODIMM	Small Outline Dual In-line Memory Module	A compact type of RAM module used mainly in laptops, small-form-factor PCs, and some all-in-one systems.
156	SOHO	Small Office/Home Office	Small Office/Home Office, refers to small-scale business/home networks and equipment.
157	SOP	Standard Operating Procedure	A documented set of step-by-step instructions that outlines how to perform routine tasks consistently and safely within an organization.
158	SPF	Sender Policy Framework	Sender Policy Framework, DNS-based email authentication to verify which mail servers can send for a domain; helps stop spoofing.
159	SPICE	Software Process Improvement and Capability Determination (1201)	A framework (ISO/IEC 15504) used to assess and improve software development processes within organizations.
160	SPICE	Simple Protocol for Independent Computing Environments (1202)	A remote-display protocol that delivers virtual desktops and applications from servers to client devices, commonly used with virtualization platforms like KVM.
161	SQL	Structured Query Language	Structured Query Language, language to create, read, update, and delete data in relational databases.
162	SSD	Solid-state Drive	Solid-State Drive, flash-based storage with fast read/write speeds and no moving parts.
163	SSH	Secure Shell	Secure Shell, encrypted remote login and command execution protocol for administering systems.
164	SSID	Service Set Identifier	Service Set Identifier, the broadcast name of a Wi-Fi network that clients see when connecting.
165	SSO	Single Sign-on	Single Sign-On, lets a user authenticate once to access multiple apps/services without re-entering credentials.
166	ST	Straight Tip	Straight Tip, bayonet-style fiber-optic connector commonly used in older networks and instrumentation.

167	TACACS	Terminal Access Controller Access-control System	A protocol used to provide centralized authentication, authorization, and accounting for users accessing network devices.
168	TCP	Transmission Control Protocol	A core protocol in the Internet Protocol Suite that ensures reliable, ordered, and error-checked delivery of data between applications.
169	TKIP	Temporal Key Integrity Protocol	An older Wi-Fi encryption protocol used with WPA that dynamically changes encryption keys to improve wireless security.
170	TN	Twisted Nematic	A type of LCD panel technology known for fast response times, commonly used in monitors and displays.
171	TOTP	Time-based One-time Password	A temporary, time-based password used for multi-factor authentication; it changes every set number of seconds.
172	TPM	Trusted Platform Module	A tamper-resistant chip on the motherboard that securely stores crypto keys and measures boot integrity (used for BitLocker, secure boot, and device attestation).
173	TXT	Text	Plain-text file format/extension with no special formatting. <i>Also in DNS, a TXT record stores arbitrary text like SPF/DMARC data.</i>
174	UAC	User Account Control	Windows feature that prompts before allowing administrative actions, reducing the risk of malware or accidental system changes.
175	UDP	User Datagram Protocol	A transport layer protocol that sends data without connection setup or delivery guarantees, used for streaming and gaming.
176	UEFI	Unified Extensible Firmware Interface	Modern PC firmware that initializes hardware and boots the OS; supports larger disks, faster boot, Secure Boot, and a graphical setup.
177	UPnP	Universal Plug and Play	Network protocol that lets devices auto-discover and request port forwarding on a router; convenient but can be a security risk if left enabled on internet-facing networks.
178	UPS	Uninterruptible Power Supply	A backup power device that supplies electricity to computers or equipment during short power outages or fluctuations.
179	USB	Universal Serial Bus	A standard interface used to connect peripherals like keyboards, mice, and storage devices to computers.
180	USB-C	Universal Serial Bus Type C	Reversible USB connector that can carry data, video, and power; supports USB 3.x/4, DisplayPort Alt Mode, and USB Power Delivery.
181	UTM	Unified Threat Management	All-in-one security appliance (often a next-gen firewall) combining functions like firewalling, IDS/IPS, AV, web filtering, and VPN.
182	VA	Vertical Alignment	LCD panel technology known for high contrast and deep blacks, typically with slower response times than TN/IPS.
183	VDI	Virtual Desktop Infrastructure	Centralized desktops run on servers and are delivered to users remotely, simplifying management and improving security.
184	VGA	Video Graphics Array	An older analog display connector standard used to connect monitors to computers.
185	VLAN	Virtual LAN [Local Area Network]	Logically segments a switch/network into separate broadcast domains for isolation and better performance without extra physical switches.
186	VM	Virtual Machine	A software-emulated computer running its own OS and apps on a host via a hypervisor, isolating workloads.

187	VNC	Virtual Network Computer (Computing?)	Remote-desktop protocol that shares a graphical desktop over the network for control and support.
188	VoIP	Voice over Internet Protocol	Technology that carries phone calls over IP networks instead of traditional phone lines.
189	VPN	Virtual Private Network	A secure network connection that encrypts internet traffic and allows private communication over public networks.
190	VRAM	Video Random-access Memory	A type of memory used by a computer's graphics card to store image and video data for display rendering.
191	WAN	Wide Area Network	A wide area network that connects computers and networks over large geographic distances, such as the internet.
192	WAP	Wireless Access Point	A wireless access point that connects wireless devices to a wired network, enabling Wi-Fi connectivity.
193	WEP	Wired Equivalency Privacy	Outdated and insecure wireless encryption standard used to protect Wi-Fi networks. It has known vulnerabilities and should not be used, replaced by WPA2/WPA3.
194	WinRM	Windows Remote Management	Microsoft's remote-management service using WS-Management; enables PowerShell remoting over HTTP/HTTPS (default ports 5985/5986).
195	WISP	Wireless Internet Service Provider	An ISP that delivers internet access via wireless (often fixed wireless) links.
196	WLAN	Wireless LAN [Local Area Network]	A local area network that uses wireless communication to connect devices within a limited area like a home or office.
197	WPA	Wi-Fi Protected Access	A Wi-Fi security protocol that improves encryption and authentication compared to the older WEP standard.
198	WWAN	Wireless Wide Area Network	Cellular-based data networking (e.g., LTE/5G) for wide-area connectivity beyond local Wi-Fi.
199	XaaS	Anything As A Service	An umbrella term for cloud-based services delivered over the internet, such as SaaS, PaaS, and IaaS.
200	XDR	Extended Detection and Response	Security platform that correlates telemetry across endpoints, network, email, and cloud to detect and respond to threats faster.
201	XFS	Extent File System	High-performance, 64-bit journaling file system (commonly on Linux) optimized for very large files and scalability.
202	XSS	Cross-site Scripting	A web attack where malicious scripts are injected into trusted sites; can steal cookies, sessions, or perform actions as the user.